
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

Anonymous authors
Anonymous submission

ABSTRACT

Algorithms increasingly serve as information mediators – from social media feeds and targeted advertising to the increasing ubiquity of LLMs – creating a joint process in which agents combine private, algorithmically-mediated signals with observational learning from their peers. This paper makes three contributions to the study of such settings. *First, as a modeling contribution*, we introduce controlled sequential social learning: a framework in which an information-mediating planner (e.g., an LLM) sets the precision of agents’ private signals while those agents also learn from the decisions of earlier agents. The planner may be *altruistic* (seeking to improve social welfare) or *biased* (seeking to induce a specific action it prefers), and its problem defines a new optimization for social learning that couples dynamic programming with decentralized action choices and Bayesian belief updates. *Second, as a theoretical contribution*, we prove that the altruistic value function is convex in the public belief and, using this, we fully characterize the optimal policies of both planners as they trade off the cost of intervention against the payoff from induced agent choices. This characterization is our central analytical result: the optimal planner operates in distinct modes across belief ranges – investing the maximum allowed resource, investing nothing, or scaling investment up or down with the belief – and, strikingly, for some belief ranges the biased planner *intentionally obfuscates* the agents’ signals. A further consequence we highlight is that, even under stringent transparency constraints – information parity with individuals, no lying or cherry-picking, and full observability – information mediation can substantially shift social welfare in either direction. *Third, as an empirical contribution*, we complement the theory with simulations in which LLMs act as both planner and agents. Here our key finding is that the LLM planner exhibits emergent strategic behavior in steering public opinion that broadly mirrors the predicted trends, while key deviations reveal the influence of non-Bayesian reasoning – consistent with the cognitive patterns of both human users and LLMs trained on human-like data. Together, these contributions establish our framework as a tractable basis for studying and regulating LLM information mediators in a way that corresponds to real behavior.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly deployed as information mediators in socially-critical functions, from search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) to personal assistants offering medical or political advice (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). While promising, LLMs have also demonstrated the ability to be persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) in their user interactions. The central premise of this paper is that, when deployed at scale, this algorithmic influence does not occur in a vacuum: it intersects with the organic spread of information as people observe and learn from one another, creating complex social dynamics that can magnify or mitigate the models’ impact. Our goal is to make that intersection analytically tractable, and our contributions – a formal model, a full characterization of optimal mediation policies, and an LLM-based empirical validation – follow directly from taking it seriously.

To make this concrete, consider our running example: an LLM-powered recommendation system that suggests a new restaurant to a sequence of potential customers. The quality of the restaurant—the true state of the world—is either good or bad, but this is unknown to the users and the system alike. Each user sees a personalized recommendation (a private signal) and then decides whether to patronize the restaurant. The recommendation system (the *planner*) can invest resources to make its signal more informative; for example, by having the LLM generate a highly tailored review based on the user’s known preferences. This investment represents a realistic cost, corresponding to expenditures on market research, user surveys, or data acquisition needed to tailor information effectively. This example fixes the two ingredients that our framework must jointly model: a planner who controls signal precision, and users who do not decide in isolation.

Crucially, however, a user’s decision is not based solely on this private signal. They also observe the actions of their predecessors—if the last ten users decided to patronize the restaurant, the next user will infer that they likely received positive information. This phenomenon, known as *social learning*, is a powerful force in shaping human opinion in a variety of settings including the adoption of vaccines (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), new technology (e.g. Gillingham & Bollinger (2021); Weber (2012)), and political and moral opinions (e.g., Brady et al. (2021); Guilbeault et al. (2018)). This is precisely the interaction our work sets out to formalize: the strategic coupling between a planner who controls private information and a population of agents who engage in social learning. The key conceptual move – and the reason the planner’s problem is nontrivial – is that the planner must anticipate how each action not only influences the current user but also creates an information *externality* that shapes the public belief confronting all future users.

While real-world interactions are immensely complex, our first contribution is to distill this dynamic into a tractable formal model of *controlled social learning*, in which an algorithmic planner strategically chooses the precision of private signals for a sequence of agents at some cost. The influence this affords the planner is deliberately subtle: it does not falsify information, but only decides how much to invest in making its signals clear and informative. Framed this way, the model lets us pose – and, in the sections that follow, answer – three critical questions that arise from this new paradigm: How might a planner wield its power to steer collective beliefs? How should its strategy adapt to evolving public opinion? And what are the ultimate impacts on social welfare?

The answers depend critically on the planner’s objective, which is why we study two objectives throughout the paper. An *altruistic* planner, like an ideal recommendation system, seeks to maximize user utility by helping them make the correct choice (patronize the restaurant if and only if it is good). In contrast, a *biased* planner, perhaps receiving a kickback from the restaurant, wants to induce a specific action (patronize) regardless of the true state. Treating both objectives on equal footing is what lets our characterization speak to information mediation as a force for good and for harm.

This altruistic/biased distinction is different from, though related to, the concept of alignment. An altruistic planner’s objective is, by definition, aligned with maximizing the agent’s *expected utility* under state uncertainty. A biased planner’s objective, being state-independent, is not; however, this does not mean its actions are always detrimental. If the true state happens to favor the planner’s preferred action (e.g., the restaurant is indeed good), its influence becomes aligned with the agent’s *realized utility* in that specific instance. We deliberately explore the full spectrum of these interactions, examining both altruistic and biased planners in scenarios where their goals may be aligned or misaligned with that of the user.

Contributions and Outline In service of the questions above, this paper makes three principal contributions, one modeling, one theoretical, and one empirical:

1. **A novel theoretical framework for controlled social learning (modeling contribution; Section 3).** We introduce the first formal model that integrates a dynamic control problem for a centralized information planner with the mechanism of sequential social learning. The planner strategically chooses the precision of agents’ private signals to achieve an objective that may be altruistic (maximizing social welfare) or biased (inducing a specific action). The novelty is the coupling: signal precision today reshapes the public belief that all future agents inherit.
2. **A rigorous characterization of optimal planner policies (theoretical contribution; Sections 4–5).** We characterize the optimal policies for both altruistic and biased planners as a function of the evolving public belief. The characterization rests on a novel proof that

the altruistic value function is convex, which may be of independent interest. Concretely, these results expose the strategic trade-offs a planner faces when its actions create informational externalities – including the striking regime in which a biased planner *deliberately degrades* signal precision to obfuscate the true state.

3. **Empirical validation and strategic analysis using LLMs (empirical contribution; Section 6).** We conduct simulations in which LLMs act as both planner and agents, yielding three concrete findings: (a) a planner that accounts for social learning can influence public opinion and social welfare far more than a myopic one; (b) the strategy that emerges from the LLM planner closely tracks our (often non-obvious) theoretical predictions, indicating the characterization is robust to non-Bayesian agent behavior; and (c) LLMs exhibit sophisticated strategic reasoning, underscoring both their potential and the societal risks they pose.

The remainder of the paper is organized around these contributions. Section 2 situates them relative to prior work. Section 3 develops the framework (Contribution 1). Sections 4 and 5 derive the optimal altruistic and biased policies (Contribution 2). Section 6 presents the LLM-based validation (Contribution 3), and Section 7 concludes. Proofs and additional results are relegated to the appendices.

2 RELATED WORK

Our contributions sit at the confluence of four literatures – social learning, information design, online persuasion and RL, and LLMs – and are best understood by what distinguishes them from each. We take each in turn, in every case highlighting the gap our framework fills.

Social Learning The sequential social learning framework we build on originates with Bikhchandani et al. (1992) and Banerjee (1992), and subsequent work has thoroughly characterized its dynamics in more general settings (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)), including many variations in the social information available to agents and in the biases and misperceptions agents may hold (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010) as well as Bikhchandani et al. (2024) and the references therein). What this line of work largely takes as fixed – the information structure feeding the agents – is exactly what our planner controls.

The literature that does study the *control* of social learning, and is therefore closest to our modeling contribution, remains relatively limited (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The two nearest works are Wei & Anastasopoulos (2022) and Smith et al. (2021). Wei & Anastasopoulos (2022) assumes two-way communication between the planner and agents (agents report their private signals to the planner in exchange for action recommendations), while Smith et al. (2021) considers a benevolent planner who directly alters agents’ choice rules and derives a decentralized pivot mechanism to improve learning. Our framework departs from both on the axis that matters for algorithmic mediation: it requires no two-way communication and leaves agents in full control of their actions, making it a better fit for information-mediating algorithms that are often invisible or black-boxed to users.

Information Design Our planner optimizes its objective by choosing the agents’ private-signal precision, which makes each period a constrained instance of classical information design (Bergemann & Morris, 2019). These models rose to prominence with Kamenica & Gentzkow (2011)’s Bayesian persuasion, in which a planner chooses the signal structure informing a rational Bayesian agent about an unknown state. Our per-period problem is a costly, constrained version of this: the planner selects a binary, symmetric signal at a cost. Against the many variations of information design, the distinguishing feature of our framework – and the source of its difficulty – is that we embed social learning *between* the agents.

Only a few works consider information design within sequential social learning (Arieli et al., 2022; Wu et al., 2025), and contrasting with them sharpens our theoretical contribution. Both study a one-shot setting in which the sender commits to a single information structure that is then fixed across the entire sequence of agents: Arieli et al. (2022) considers a self-interested sender choosing a general structure as in classic Bayesian persuasion, while Wu et al. (2025) constrain the structure to be Blackwell-more-informative and maximize the probability of a correct cascade. We instead solve a genuinely dynamic problem in which the sender chooses a fresh structure for every agent,

giving rise to a substantially richer class of sender strategies – the very class our characterization is designed to describe.

Online Persuasion and RL A growing body of work sits at the interface of online learning and Bayesian persuasion/information design, and it is worth delineating precisely because our sequential setting could be mistaken for it. One strand studies online sequential persuasion under unknown environment parameters, where a sender interacts repeatedly with short-lived receivers while lacking knowledge of the prior over states and/or receivers’ utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A related strand examines sequential persuasion in dynamic environments or MDPs, introducing state transitions and/or receiver foresight; these emphasize history-dependent or promise-based signaling policies and often connect to reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). Across both strands, the goal is learning-based performance guarantees – sublinear regret against the best signaling scheme in hindsight, or approximation algorithms under computational constraints.

Our contribution is orthogonal to these goals. The planner in our model does not learn unknown parameters, and it is not informationally advantaged – it observes neither the underlying state nor the private signals. Instead of arising from an uninformed sender exploring an unknown environment, the sequential dependence in our setting arises entirely from *social learning among the receivers*: agents’ actions shape the beliefs of future agents. Concretely, ours is not an online-learning or RL environment, because actions do not modify the underlying payoff-relevant state (there are no reward externalities) and the planner learns only by observing agents’ actions rather than by payoff exploration or state evolution. It is precisely because the environment is known and stationary that we can go beyond regret bounds and instead characterize the planner’s optimal policies exactly.

LLMs Most recently, several works have studied LLMs specifically in the context of information design (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). Crucially for our empirical contribution, all of these consider one-on-one interactions and thus do not capture the social dynamics we investigate. The closest is Duetting et al. (2025), who study information design where a sender chooses both a signal structure and a ‘framing’ that conditions the (possibly non-Bayesian) posterior of the receiver, and who implement an LLM-based oracle to optimize over the computationally difficult framing space – building on the notion that LLMs can mimic humans as economic agents (Dillion et al., 2023; Horton, 2023). Their work does not address the repeated interactions or social learning at the heart of our study; we do, however, adapt parts of their oracle methodology for the LLM experiments of Section 6.

3 SOCIAL LEARNING MODEL

This section develops our first contribution: a formal model of controlled social learning. Its defining feature, which we build up component by component below, is that a single planner shapes the private information of every agent while those agents simultaneously learn from one another. We consider a planner and a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature determines a fixed, *unknown* exogenous state of the world $\omega \in \Omega := \{G, B\}$, where $\mathbb{P}(\omega = G) = b_1$, and b_1 is known to everyone. In the recommendation system example of Section 1, the state G corresponds to the case where the restaurant is good.

At each time i , the planner decides whether or not to invest in personalization for agent i . Subsequently, it provides agent i with a private signal $s_i \in \Omega$ to aid in decision-making. In our model, s_i is assumed binary and it matches the state of the world ω with probability $q_i \in [0.5, 1]$ i.e. $\mathbb{P}(s_i = \omega) = q_i$. We refer to q_i as agent i ’s signal precision.

Each agent’s signal is independent of those of other agents and the history when conditioned on ω . Based on this signal and the observed actions of previous agents $j < i$, agent i selects an action $a_i \in \Omega$. The agent receives utility 0 if her action matches the true state ($a_i = \omega$), and utility $-C$ otherwise, where $C > 0$. Each player chooses her action to maximize her own expected utility. For example, this corresponds to patronizing a business if and only if the state is G .

3.1 AGENTS' DECISION PROBLEMS

The single state variable that will carry the entire analysis is the public belief, which we now define. Each agent observes the actions of all her predecessors and their respective signal precisions, a history denoted $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because all agents and the planner share the identical information $\mathcal{H}_i$, they also share a *public* belief about ω that is updated after each agent acts. The public belief b_i just before agent i acts is $\mathbb{P}(\omega = G | \mathcal{H}_i)$; before any agent acts it is the prior b_1 . As in the classic model of Banerjee (1992) and Bikhchandani et al. (1992), $(b_i)_{i \in \mathbb{N}}$ is a Markov process and a sufficient statistic for the history $\mathcal{H}_i$ (proof in Appendix B.3). This sufficiency is what makes the planner's problem a tractable, low-dimensional control problem rather than one over unwieldy histories.

We now specify how an agent acts, since it is the agent's action – not her signal – that becomes public and drives the learning dynamics. Informed by $\mathcal{H}_i$ (equivalently, by the Markov property, b_i) and q_i , agent i chooses the action $a_i \in \Omega$ that maximizes her utility, breaking ties in favor of the action matching s_i . She first forms a *private* belief $\tilde{b}_i$ about ω from $\mathcal{H}_i$ and her private signal s_i of precision q_i as follows (derivation in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

Using this private belief, she selects whichever state is more likely under $\tilde{b}_i$: $a_i = G$ if $\tilde{b}_i > 0.5$, $a_i = B$ if $\tilde{b}_i < 0.5$, and $a_i = s_i$ if $\tilde{b}_i = 0.5$. Substituting for $\tilde{b}_i$ yields the action rule that we will invoke repeatedly throughout the characterization:

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Agent i 's action is then observed by subsequent agents and incorporated into the updated public belief b_{i+1} as follows (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. *The update in Equation (3) exposes the central tension the planner must manage. When $1 - q_i \leq b_i \leq q_i$, agent i 's action perfectly reveals her private signal via Equation (2), so the public belief moves to her private belief and society learns. Otherwise, Equation (2) makes agent i 's private signal uninformative for her action, the public belief is frozen, and an absorbing state – an information cascade or herding, in the language of Bikhchandani et al. (1992); Banerjee (1992) – is reached, at which point society stops learning from agents' private information. Steering the system into or out of such cascades is exactly the lever our planner exploits.*

Agent i 's expected utility is $-C \mathbb{P}(a_i \neq \omega | b_i, q_i)$ which has the following form (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

Having fixed the agents' behavior, we now introduce the two planners whose optimal policies are the subject of our theoretical contribution: (1) an *altruistic* planner, who wishes to induce the correct action ($a_i = \omega$), and (2) a *biased* planner, who wishes to induce a fixed action, say G , regardless of ω . We denote them with subscripts A and B , respectively. In both cases the planner sets each agent's signal precision at a cost given by a function $\beta(\cdot)$ that is non-negative, increasing, continuous, and concave in its argument, and – importantly for the transparency claims we make later – the planner's information set $\mathcal{H}_i$ is identical to the agents'.

Altruistic Planner At time i , the altruistic planner chooses the precision q_i for agent i and incurs cost $\beta(q_i)$ with $\beta(p) = 0$ for a baseline $p \in [0.5, 1)$. Because decreasing precision never benefits the altruistic planner (see Appendix C.9), we simplify notation by charging the planner only for

raising precision above the baseline p , with the extra cost rising – at a decreasing marginal rate – as precision increases further. Both p and $\beta(\cdot)$ are known to the agents.

The altruistic planner’s objective is to maximize social welfare – the expected total utility of the agents – net of its precision investment. Writing $r_A(b_i, q_i)$ for the instantaneous reward of an altruistic planner that begins at public belief b_i and sets precision q_i for agent i , the discounted objective it maximizes, starting from b_1 and following a policy sequence $\pi = (\pi_i)_{i=1}^\infty$ with $\pi_i(\mathcal{H}_i) = q_i$ and discount factor $\delta \in [0, 1)$, is:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The altruistic planner’s optimal utility and policy are the supremum and arg supremum of $V_A^\pi(\cdot)$ over the policy set Π . A useful special case that we return to as a baseline throughout the analysis is the *myopic* planner, recovered by setting $\delta = 0$ so that all future consequences of social learning are ignored.

Biased Planner The biased planner’s problem differs from the altruistic one only in its objective, and hence in its cost and reward functions; the political-campaign example of Appendix A illustrates the biased case in full. This planner seeks action G from every agent regardless of ω : it incurs cost 0 when an agent chooses G and cost $C > 0$ when an agent chooses B , independent of the true state. The key structural difference is that the biased planner may push precision either above or below the baseline p , and either direction is costly, since any deviation from p requires researching and tailoring the ad to an individual agent. Concretely, choosing precision q_i costs $\beta(|q_i - p|)$, with $\beta(0) = 0$.

This gives the biased planner’s expected instantaneous reward at time i as:

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

The biased planner’s value functions $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and optimal policy $\pi_B^*(\cdot)$ are defined from Equation (6) exactly as their altruistic counterparts were defined from Equation (5).

It is worth pausing on what separates the two objectives, because this distinction organizes all of our welfare findings. Since each agent still prefers to match ω , a biased planner that succeeds when $\omega = B$ actively lowers the agent’s utility, whereas an altruistic planner – who wants $a_i = \omega$ whatever ω is – always raises it. In short, the altruistic planner’s objective is always aligned with each agent’s, while the biased planner’s is aligned only when $\omega = G$.

Both objectives thus reduce to infinite-horizon discounted stationary Markov Decision Processes (MDPs) with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and the transition function of Equation (3) (Puterman, 1990), each admitting a unique optimal value function (Kallenberg, 2011). This MDP reduction, enabled by the sufficiency of the public belief, is what makes the exact characterization in Sections 4–5 possible. Throughout, we restrict attention to deterministic Markov policies, so $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$.

Remark 2. *We make three assumptions worth flagging explicitly, as they are precisely the transparency constraints that make our welfare results conservative – any harm we exhibit occurs despite them. (1) The planner has the same history as the agents: limiting when the planner could hold richer data, but innocuous when all parties rely on the same public signals, such as early findings or survey data. (2) The planner can change agents’ signal precision, but every signal still passes through a binary symmetric channel: limiting if planners can cherry-pick, censor, or falsify content, but innocuous when their role is only to improve the quality of noisy sources without altering content. (3) The planner’s control choices are fully observable: limiting when covert policies or framing are possible, but innocuous in more transparent environments.*

4 OPTIMAL ALTRUISTIC POLICIES

This section delivers the first half of our theoretical contribution: a complete characterization of the optimal altruistic policy. We build to it in three steps – the myopic benchmark, the convexity of the value function, and the full optimal policy – so that the role of social learning is visible at each stage.

We begin with the myopic case, which disregards social learning by setting $\delta = 0$ so the planner ignores all future costs; then $V_A^\pi(b) = r_A(b, \pi(b))$ and the optimal myopic policy $\pi_A^0(\cdot)$ is:

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

This myopic problem has an instructive equivalent reading: it is exactly the decentralized case in which each agent chooses the precision of her own signal and bears the cost, maximizing her own expected utility net of cost. The gap between this benchmark and the forward-looking optimum below is precisely the value of accounting for social learning.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimum is therefore a simple threshold policy: at strong public beliefs the planner spends nothing and uses the baseline precision p , while at weak beliefs it supplies a perfect signal (precision 1). The threshold depends only on the cost of a perfect signal ($\beta(1)$) relative to the cost of an incorrect action (C): once $\beta(1) \geq C$, the perfect signal is too expensive ever to use, so $t_M = 0.5$ and the interval of beliefs at which precision 1 is optimal vanishes. Keep this threshold shape in mind, as the forward-looking optimum below both preserves and strictly enlarges it.

We now turn to the forward-looking planner. The cornerstone of our characterization – and the technical heart of the theoretical contribution – is the following structural result for the altruistic value function:

Theorem 2. *$V_A^*(\cdot)$ is convex with respect to public belief.*

The proof of Theorem 2 (Appendix C.3) is quite involved and, we believe, of independent interest. Its difficulty stems from the very coupling that defines our model: agents' actions depend on the public belief. Were actions independent of the belief process (as in Nyarko (1994)), expected utility would be linear in the belief and convexity would follow immediately; here it does not, and establishing it is a genuine contribution in its own right.

With convexity in hand, we can now state our first main characterization result, whose three-phase structure is the payoff of all the machinery above.

Theorem 3. *There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and*

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & \text{o.w.} \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

Theorem 3 says the optimal altruistic policy has three distinct phases in the public belief, and it is instructive to read each against the myopic benchmark. *First*, like the myopic optimum, the forward-looking optimum stops investing at extreme beliefs – but it requires a *stronger* belief before doing so, precisely because it, unlike the myopic planner, accounts for the effect on future agents.

Second, when the public belief is near 0.5 and carries little information, the optimum supplies a perfect signal (precision 1) whenever this is not cost-prohibitive; by Equation (2) the agent's action then matches the true state with probability 1, collapsing the public belief to 0 or 1.

Third, in the intermediate range the optimum chooses the smallest precision $\max(b, 1 - b)$ that still lets the agent's action reflect her private signal (see Equation (2)). Any lower precision would make the action redundant with what peers already know, so this is exactly the cheapest precision that keeps social learning alive through observation of peers.

5 OPTIMAL BIASED POLICIES

We now complete the theoretical contribution with the biased planner, mirroring the structure of Section 4: first the myopic optimum, then the full optimal policy. This case is where the framework's

most striking prediction emerges – that a transparent, non-lying planner may nonetheless choose to *obfuscate*. As before, the myopic biased optimum $\pi_B^0(\cdot)$ is the arg supremum of the instantaneous reward $r_B(b, q)$. A new subtlety arises here that was absent in the altruistic case: an exact optimum need not exist (Puterman, 1990, sec. 2.3.1). Where this occurs we characterize ϵ -optimal policies instead, denoted with an ϵ superscript.

Theorem 4. *There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:*

- (A) *If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.*
- (B) *If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.*
- (C) *If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.*
- (D) *If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Proof in Appendix C.6. Building on this myopic result, we now state the paper’s second main characterization – the optimal biased policy.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) *If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.*
- (B) *If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.*
- (C) *If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.*
- (D) *If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.*
- (E) *If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon*}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Theorem 5 (proven in Appendix C.8 and depicted in figure 2a) reveals five distinct phases, which together tell a coherent story of how a biased mediator steers opinion. We walk through them in turn.

Phase (A): strong belief, no intervention. When the public belief is already strong, the cost of steering the system is prohibitive – even though a low belief ($b < 0.5$) hurts the planner. The chosen precision p falls below $\max(b, 1 - b)$, so by Equation (2) the agent follows the public belief regardless of her signal; the action is uninformative, the belief is frozen, and, because the policy is Markovian, this repeats forever. The result is an unfavorable cascade if $b < 0.5$ and a favorable one otherwise.

Phase (B): last-ditch effort near an unfavorable cascade. As the belief approaches an unfavorable cascade, the planner raises precision just enough to make the agent’s action responsive to her signal, even though that signal will most likely be B . This gives a nonzero chance of a favorable action – a last-ditch attempt to escape the unfavorable cascade.

Phase (C): the obfuscation regime. At slightly higher beliefs (still with $b < 0.5$), where more precise signals more often deliver bad news, the planner *lowers* precision below p – yet keeps it strong enough to move the agent’s action ($q \geq 1 - b$). In other words, it steers toward a more favorable belief with the least informative signal possible. This is the striking regime foreshadowed above: a fully transparent planner deliberately degrading information.

Phase (D): bolstering a favorable belief. When the belief weakly favors the planner ($b > 0.5$), it adopts precision at least p ; since higher precision now makes $\omega = G$ more likely to be inferred, this investment simply reinforces the favorable belief.

Phase (E): protecting a favorable belief. At still higher beliefs the planner drops precision just below $\max(b, 1 - b)$ to $b - \epsilon$, so agents ignore their signals and choose G . Here the risk that a private signal overturns the favorable belief outweighs both the cost of lowering precision and the upside of pushing the belief still higher.

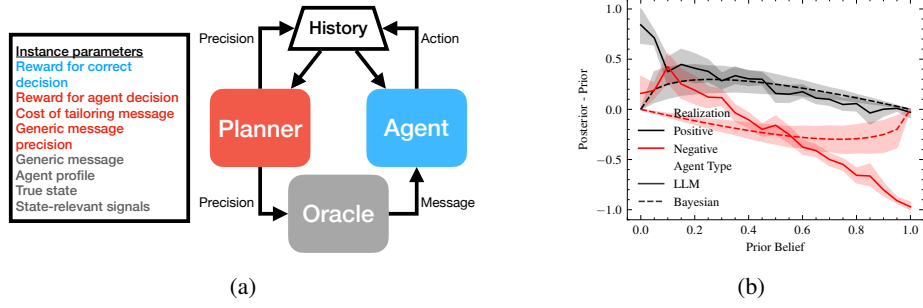


Figure 1: Overview of the empirical setup and the first key finding. (a) System diagram for our experiments: the instance parameters are color-coded by which LLM role consumes them. (b) The change in LLM (solid) and Bayesian (dashed) agents’ beliefs for a given prior after a positive (black) or negative (red) signal, illustrating the systematic non-Bayesian updating that our planner must contend with.

6 EVALUATION VIA LLM-BASED SIMULATION

We now turn to our third contribution: an empirical validation that puts the framework and its predictions to the test with LLMs acting as both planner and agents. The evaluation is structured to isolate three questions in sequence – how LLM agents deviate from Bayesian rationality, how an LLM planner adapts its strategy in response, and what the resulting impact on social welfare is – each of which speaks directly to whether our theoretical characterization survives contact with realistic, non-Bayesian behavior.

To operationalize controlled social learning, we instantiate our model in a scenario where agents decide whether to buy a new car. As shown in figure 1a, LLMs occupy three roles, described below with corresponding prompts in Appendix E.2.2:

Agent: Role-playing as an assigned profile, agents observe previous actions and a private message about the car of known precision, form their belief about the car’s quality, and decide to buy or not.

Planner: After observing the history of actions, the planner selects the precision q_i of agent i ’s private signal according to their objective

Oracle: Given the choice of precision by the planner, an agent profile, and a fact sheet about the car (see Appendix E.2.1), the oracle generates a private signal of desired precision tailored to an agent. Throughout, we compare these LLM-based simulations against numerical solutions of the planner’s MDP. Both use a linear cost function $\beta(q) = k|q - p|$, vary the cost scale k , baseline precision p , and discount factor δ , and fix the cost of an incorrect action at $C = 1$. Appendix E details the setup and prompting, and Appendix E.3 validates both the LLM beliefs and the oracle’s performance.

The experiments substantiate three findings, which correspond one-to-one to the three findings pre-viewed in our contributions and are developed in the three subsections that follow:

1. **Social-learning control matters (Section 6.3):** both analytical and LLM planners dramatically improve their own utility by accounting for and exploiting social learning – helping or harming social welfare depending on whether planner and agent objectives are aligned.
2. **Our characterization is robust (Section 6.2):** LLM planners choose policies strikingly close to the non-obvious analytically optimal ones, even while facing non-Bayesian LLM agents.
3. **LLMs are emergently strategic (Sections 6.1 and 6.2):** LLM planners not only approximate the optimal policies but also deviate from them in ways that account for the non-Bayesian tendencies of the LLM agents they face.

6.1 LLMs AS NON-BAYESIAN AGENTS

Our characterization assumes Bayesian agents, so before testing it we must first document how far LLM agents depart from that assumption – this is the necessary groundwork for Finding 3. Since

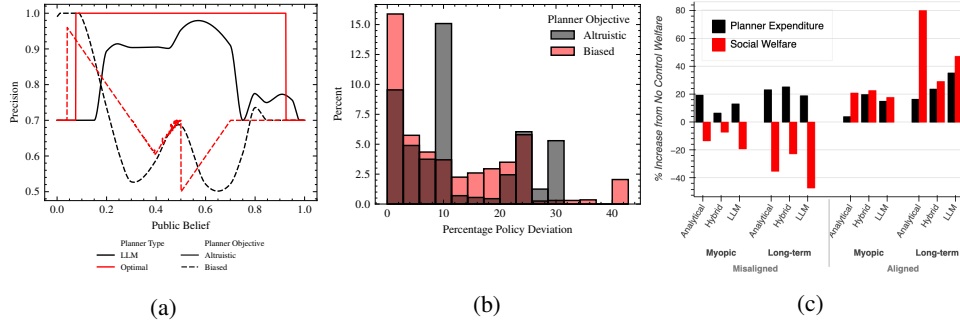


Figure 2: Empirical evidence for Findings 2 and 3. (a) Example policies from the LLM planner (black) and the analytically optimal planner (red) in altruistic (solid) and biased (dashed) settings, showing close structural agreement even for the non-obvious biased policy. (b) Distribution of the percentage deviation between the LLM and optimal policies, quantifying that agreement. (c) Planner expenditure and social welfare change as a percent of the no-control baseline welfare, with the true state fixed to B; the left half is a biased planner seeking action G and the right half an altruistic planner.

both humans and LLMs are known to exhibit systematic cognitive biases (Rupprecht et al., 2025), we isolate the LLM’s by measuring how its belief updates in response to new information. As figure 1b shows, the updates depart from Bayes’ rule in three specific and consistent ways (see also Appendix E.4):

- (NB1) LLM agents underreact to private signals which align with their prior beliefs.
- (NB2) LLM agents overreact to private signals which run counter to their prior beliefs.
- (NB3) As a result, LLM agents require a stronger public belief to enter an information cascade.

Reassuringly for the external validity of our simulations, these same biases have been documented empirically in human studies (e.g., Ba et al. (2022); Chan et al. (2025)). The subsequent sections establish our key robustness finding: the LLM planner’s policy holds up well against non-Bayesian agents of exactly this kind.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

This subsection presents the core evidence for Finding 2, the robustness of our characterization, by comparing the LLM planner’s policies against the optima derived in Sections 4 and 5. As figure 2a shows, and despite the agent-level deviations just documented, the LLM planner’s emergent strategy is remarkably close in structure to the theoretical optimum. The headline is that even when the optimal policy is highly non-trivial (e.g., the biased policy in figure 2a), the LLM planner reproduces sophisticated strategic behavior that both accounts for and exploits social learning.

The agreement is qualitative and quantitative. Qualitatively, in the altruistic case both planners stop investing at strong beliefs and invest heavily under uncertainty; in the biased case both show the same signature – heavy investment to escape an unfavorable belief, reduced precision near the midpoint, and no investment once a favorable belief is established. Quantitatively, figure 2b shows the deviation is below 10% for the majority of belief states in both settings, underscoring the broad structural alignment between the emergent LLM strategy and our characterization.

Just as informative are the structural differences, which – and this is the essence of Finding 3 – read as the LLM planner’s strategic adaptations to the specific non-Bayesian behaviors of Section 6.1. (1) The LLM planner avoids extreme precisions (0.5 or 1.0), consistent with a known central-tendency bias (Rupprecht et al., 2025). (2) It tapers investment gradually rather than cutting off sharply – a direct response to the agents’ resistance to cascades (NB3), having learned that it is never entirely “safe” to stop investing because even agents with strong priors can be swayed by a countervailing signal. (3) The biased LLM planner keeps investing at very low beliefs, reflecting an understanding that its agents may overreact (NB2) to a surprisingly positive signal, which makes the “last-ditch effort” more viable than in the Bayesian case.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

This subsection develops Finding 1 – that controlling social learning has a large, direction-dependent welfare impact. We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Because better information never lowers an agent’s expected utility, welfare is monotonic in signal precision (Appendix C.9), so an altruistic planner always raises welfare above the baseline. To quantify the effect, we compare welfare and planner expenditure across three settings: (1) *analytical* (optimal policy, Bayesian agents), (2) *LLM* (LLM planner, LLM agents), and (3) *hybrid* (optimal policy, LLM agents).

Two results stand out in figure 2c. First, planners in all three settings significantly alter welfare, and neglecting social learning (the myopic cases) substantially worsens the planner’s own outcomes – direct confirmation that social-learning control is the operative lever. Second, and more concerning, the misaligned biased planners (analytical and LLM alike) *decreased* social welfare by 40 to 50%, achieved precisely by the obfuscation mechanism our theory predicts: policies that lower signal precision to hide the true state. That this much harm is possible *despite* the harsh transparency constraints of Remark 2 is what makes the finding striking, and it substantiates the risk posed by misaligned LLM information mediators.

The comparison also exposes the cost of misspecifying the agent model. In the hybrid setting, the analytically optimal policy – built for Bayes-rational agents – proves “brittle,” underperforming when applied to non-Bayesian agents. The LLM policy, by contrast, tracks the analytical policy closely while being better tuned to the human-like biases of real agents.

7 CONCLUSION

This paper delivered three contributions toward understanding algorithmic information mediation. As our modeling contribution, we introduced a formal framework for the strategic control of social learning by an information mediator. As our theoretical contribution, we characterized the optimal altruistic and biased policies – proving en route the convexity of the altruistic value function – and showed that even a tightly constrained planner can drastically move social welfare in either direction, with accounting for social learning being integral to wielding that influence. As our empirical contribution, LLM-based simulations confirmed these insights, revealing that LLMs reason strategically in a way that mirrors our analytical results while adapting to non-Bayesian agents.

Taken together, these contributions yield three takeaways for algorithmic information mediation: (1) human-AI interaction cannot be studied in a vacuum – neglecting its interplay with forces like social learning means failing to capture its potential, positive or negative; (2) information mediators, even when exercising subtle influence, have immense power to guide or derail social learning and public opinion, motivating work on mitigating the attendant risks; and (3) modern LLMs exhibit emergent strategic behavior that both accounts for and exploits social learning and non-Bayesian cognitive biases. The latter two points underscore the need to better understand how the risks of information mediators, and LLMs in particular, might be mitigated. A limitation that qualifies our empirical contribution is the absence of human data: since the fidelity of LLM-human simulators remains contested (see, e.g., Horton (2023); Gao et al. (2025)), human data could help verify – or even fine-tune – the LLM simulators.

Two future directions follow directly from these contributions: generalizing the model and its characterization, and mitigating the negative welfare effects of biased planners. On the first, several generalizations would relax our assumptions and bring the model closer to reality – larger state spaces, more general signal structures, more general loss functions, and heterogeneous agents. Moving beyond binary states and 0-1 losses is primarily an algebraic hurdle, and we conjecture the qualitative character of our results persists; for more general signal structures the key open challenge is preserving the convexity that underpins Theorem 2, and identifying the most general signal-structure class for which it holds is an appealing problem. As a first step toward heterogeneous agents, Appendix D already extends our altruistic characterization to agents with opposing preferences.

On the second, our welfare finding makes the question of preventing biased-planner harm pressing. A natural avenue is to design regulations or mechanisms that realign the incentives of planners and agents so as to avert the welfare losses we documented.

REFERENCES

- Shipra Agrawal, Yiding Feng, and Wei Tang. Dynamic Pricing and Learning with Bayesian Persuasion. *Advances in Neural Information Processing Systems*, 36:59273–59285, December 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/b9c2e8a0bbbed5fcfaf62856a3a719ada-Abstract-Conference.html.
- Lisa P. Argyle, Christopher A. Bail, Ethan C. Busby, Joshua R. Gubler, Thomas Howe, Christopher Rytting, Taylor Sorensen, and David Wingate. Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences*, 120(41):e2311627120, October 2023. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.2311627120. URL <https://pnas.org/doi/10.1073/pnas.2311627120>.
- Itai Arieli and Manuel Mueller-Frank. A general analysis of sequential social learning. *Mathematics of Operations Research*, 46:1235–1249, 11 2021.
- Itai Arieli, Ronen Gradwohl, and Rann Smorodinsky. Herd Design. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pp. 871–872, Boulder CO USA, July 2022. ACM. ISBN 978-1-4503-9150-4. doi: 10.1145/3490486.3538292. URL <https://dl.acm.org/doi/10.1145/3490486.3538292>.
- Cuimin Ba, J. Aislinn Bohren, and Alex Imas. Over- and Underreaction to Information. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4274617. URL <https://www.ssrn.com/abstract=4274617>. Publisher: Elsevier BV.
- Francesco Bacchiocchi, Matteo Bollini, Matteo Castiglioni, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion Without a Clue. *Advances in Neural Information Processing Systems*, 37:76404–76449, December 2024.
- Fengshuo Bai, Mingzhi Wang, Zhaowei Zhang, Boyuan Chen, Yinda Xu, Ying Wen, and Yaodong Yang. Efficient Model-agnostic Alignment via Bayesian Persuasion, May 2024.
- Abhijit V Banerjee. A simple model of herd behavior. *The Quarterly Journal of Economics*, 107(3):797–817, 1992.
- Larry M. Bartels. *Presidential primaries and the dynamics of public choice*. Princeton, N.J. : Princeton University Press, 1988.
- Chris T Bauch and Samit Bhattacharyya. Evolutionary game theory and social learning can determine how vaccine scares unfold. *PLoS computational biology*, 8(4), 2012.
- Dirk Bergemann and Stephen Morris. Information Design: A Unified Perspective. *Journal of Economic Literature*, 57(1):44–95, March 2019. ISSN 0022-0515. doi: 10.1257/jel.20181489. URL <https://www.aeaweb.org/articles?id=10.1257/jel.20181489>.
- Martino Bernasconi, Matteo Castiglioni, Alberto Marchesi, and Mirco Mutti. Persuading Farsighted Receivers in MDPs: The Power of Honesty. *Advances in Neural Information Processing Systems*, 36:14987–15014, December 2023.
- Sujay Bhatt and Vikram Krishnamurthy. Controlled Sequential Information Fusion With Social Sensors. *IEEE Transactions on Automatic Control*, 66(12):5893–5908, December 2021.
- Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.
- Sushil Bikhchandani, David Hirshleifer, Omer Tamuz, and Ivo Welch. Information Cascades and Social Learning. *Journal of Economic Literature*, 62(3):1040–1093, September 2024. ISSN 0022-0515, 2328-8175. doi: 10.1257/jel.20241472. URL <https://pubs.aeaweb.org/doi/10.1257/jel.20241472>.
- Ilai Bistriz, Nasimeh Heydaribeni, and Achilleas Anastasopoulos. Informational cascades with nonmyopic agents. *IEEE Transactions on Automatic Control*, 67, 9 2022.
- William J Brady, Killian McLoughlin, Tuan N Doan, and Molly J Crockett. How social learning amplifies moral outrage expression in online social networks. *Science Advances*, 7(33), 2021.
- Carlos Carrasco-Farre. Large Language Models are as persuasive as humans, but how? About the cognitive effort and moral-emotional language of LLM arguments, April 2024. URL <http://arxiv.org/abs/2404.09329>. arXiv:2404.09329 [cs].

- Matteo Castiglioni, Andrea Celli, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion. In *Advances in Neural Information Processing Systems*, volume 33, pp. 16188–16198. Curran Associates, Inc., 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/ba5451d3c91a0f982f103cdbc249bc78-Abstract.html>.
- Matteo Castiglioni, Alberto Marchesi, Andrea Celli, and Nicola Gatti. Multi-Receiver Online Bayesian Persuasion. In *Proceedings of the 38th International Conference on Machine Learning*, pp. 1314–1323. PMLR, July 2021.
- Kenneth Chan, Gary Charness, Chetan Dave, and J. Lucas Reddinger. On Prior Confidence and Belief Updating, May 2025. URL <http://arxiv.org/abs/2412.10662>. arXiv:2412.10662 [econ].
- Zirui Cheng and Jiaxuan You. Towards Strategic Persuasion with Language Models, September 2025.
- Krishna Dasaratha and Kevin He. Learning from viral content. *arXiv preprint arXiv:2210.01267*, 2022.
- Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can AI language models replace human participants? *Trends in Cognitive Sciences*, 27(7):597–600, July 2023. ISSN 1364-6613, 1879-307X. doi: 10.1016/j.tics.2023.04.008. URL [https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613\(23\)00098-0](https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613(23)00098-0). Publisher: Elsevier.
- Paul Duetting, Safwan Hossain, Tao Lin, Renato Paes Leme, Sai Srivatsa Ravindranath, Haifeng Xu, and Song Zuo. Information Design With Large Language Models, September 2025. URL <http://arxiv.org/abs/2509.25565>. arXiv:2509.25565 [cs].
- Maxim Egorov, Zachary N. Sunberg, Edward Balaban, Tim A. Wheeler, Jayesh K. Gupta, and Mykel J. Kochenderfer. POMDPs.jl: A framework for sequential decision making under uncertainty. *Journal of Machine Learning Research*, 18(26):1–5, 2017.
- Erik Eyster and Matthew Rabin. Naïve herding in rich-information settings. *American Economic Journal: Microeconomics*, 2:221–243, 11 2010.
- Shen Gao, Jiabao Fang, Quan Tu, Zhitao Yao, Zhumin Chen, Pengjie Ren, and Zhaochun Ren. Generative News Recommendation. In *Proceedings of the ACM Web Conference 2024*, WWW ’24, pp. 3444–3453, New York, NY, USA, May 2024. Association for Computing Machinery. ISBN 979-8-4007-0171-9. doi: 10.1145/3589334.3645448. URL <https://dl.acm.org/doi/10.1145/3589334.3645448>.
- Yuan Gao, Dokyun Lee, Gordon Burtch, and Sina Fazelpour. Take Caution in Using LLMs as Human Surrogates: Scylla Ex Machina, January 2025. URL <http://arxiv.org/abs/2410.19599>. arXiv:2410.19599 [econ].
- Kenneth T Gillingham and Bryan Bollinger. Social learning and solar photovoltaic adoption. *Management Science*, 67(11):7091–7112, 2021.
- Douglas Guilbeault, Joshua Becker, and Damon Centola. Social learning and partisan bias in the interpretation of climate trends. *Proceedings of the National Academy of Sciences*, 115(39):9714–9719, 2018.
- Keegan Harris, Nicole Immorlica, Brendan Lucier, and Aleksandrs Slivkins. Algorithmic Persuasion Through Simulation, February 2025. URL <http://arxiv.org/abs/2311.18138>. arXiv:2311.18138 [cs].
- Claudia E. Haupt and Mason Marks. AI-Generated Medical Advice—GPT and Beyond. *JAMA*, 329(16): 1349–1350, April 2023. ISSN 0098-7484. doi: 10.1001/jama.2023.5321. URL <https://doi.org/10.1001/jama.2023.5321>.
- K. Hinderer. Lipschitz Continuity of Value Functions in Markovian Decision Processes. *Mathematical Methods of Operations Research*, 62(1):3–22, September 2005.
- John J. Horton. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?, April 2023. URL <https://www.nber.org/papers/w31122>.
- Bright Huo, Amy Boyle, Nana Marfo, Wimonchat Tangamornsuksan, Jeremy P. Steen, Tyler McKechnie, Yung Lee, Julio Mayol, Stavros A. Antoniou, Arun James Thirunavukarasu, Stephanie Sanger, Karim Ramji, and Gordon Guyatt. Large Language Models for Chatbot Health Advice Studies: A Systematic Review. *JAMA Network Open*, 8(2):e2457879, February 2025. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2024.57879. URL <https://doi.org/10.1001/jamanetworkopen.2024.57879>.
- Cameron R. Jones and Benjamin K. Bergen. Lies, Damned Lies, and Distributional Language Statistics: Persuasion and Deception with Large Language Models, December 2024. URL <http://arxiv.org/abs/2412.17128>. arXiv:2412.17128 [cs].

-
- Lodewijk Kallenberg. Markov decision processes. *Lecture Notes. University of Leiden*, 428, 2011.
- Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.
- Vikram Krishnamurthy. Quickest Detection POMDPs With Social Learning: Interaction of Local and Global Decision Makers. *IEEE Transactions on Information Theory*, 58(8):5563–5587, August 2012.
- LG AI Research. Exaone deep: Reasoning enhanced language models. *arXiv preprint arXiv:2503.12524*, 2025.
- Minqian Liu, Zhiyang Xu, Xinyi Zhang, Heajun An, Sarvech Qadir, Qi Zhang, Pamela J. Wisniewski, Jin-Hee Cho, Sang Won Lee, Ruoxi Jia, and Lifu Huang. LLM Can be a Dangerous Persuader: Empirical Study of Persuasion Safety in Large Language Models, April 2025. URL <http://arxiv.org/abs/2504.10430>. arXiv:2504.10430 [cs].
- Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14(1), August 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-69032-z. URL <https://www.nature.com/articles/s41598-024-69032-z>. Publisher: Springer Science and Business Media LLC.
- Yaw Nyarko. On the convexity of the value function in Bayesian optimal control problems. *Economic Theory*, 4(2):303–309, March 1994.
- Kenneth Payne and Baptiste Alloui-Cros. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory, July 2025. URL <http://arxiv.org/abs/2507.02618>. arXiv:2507.02618 [cs].
- Yujin Potter, Shiyang Lai, Junsol Kim, James Evans, and Dawn Song. Hidden Persuaders: LLMs’ Political Leaning and Their Influence on Voters, November 2024.
- Martin L. Puterman. Chapter 8 markov decision processes. In *Stochastic Models*, volume 2 of *Handbooks in Operations Research and Management Science*. Elsevier, 1990.
- Maya Raifer, Guy Rotman, Reut Apel, Moshe Tennenholtz, and Roi Reichart. Designing an Automatic Agent for Repeated Language-based Persuasion Games. *Transactions of the Association for Computational Linguistics*, 10:307–324, March 2022.
- Neel Rao, Markus M Mobius, and Tanya Rosenblat. Social networks and vaccination decisions. *FRB of Boston*, 2007.
- Alexander Rogiers, Sander Noels, Maarten Buyt, and Tijl De Bie. Persuasion with Large Language Models: a Survey, November 2024. URL <http://arxiv.org/abs/2411.06837>. arXiv:2411.06837 [cs].
- Jens Rupperecht, Georg Ahnert, and Markus Strohmaier. Prompt Perturbations Reveal Human-Like Biases in LLM Survey Responses, July 2025. URL <http://arxiv.org/abs/2507.07188>. arXiv:2507.07188 [cs].
- Dr.-Ing. Martin Schiele, Yannick Gittmann, Stefan Ilchmann, Ante Gojsalić, Dominik Jurinčić, and Phylis Klempt. Voting Advice Applications: Implementation of RAG-supported LLMs, July 2024. URL <https://www.techrxiv.org/users/802685/articles/1187651-voting-advice-applications-implementation-of-rag-supported-llms?commit=c540a54749906f8ac761093f5dff0f514291d4b6>.
- James E. Smith and Kevin F. McCardle. Structural Properties of Stochastic Dynamic Programs. *Operations Research*, 50(5):796–809, October 2002.
- Lones Smith and Peter Sørensen. Pathological outcomes of observational learning. *Econometrica*, 68:371–398, 2000.
- Lones Smith, Peter Norman Sørensen, and Jianrong Tian. Informational Herding, Optimal Experimentation, and Contrarianism. *The Review of Economic Studies*, 88(5):2527–2554, September 2021. ISSN 0034-6527, 1467-937X. doi: 10.1093/restud/rdab001. URL <https://academic.oup.com/restud/article/88/5/2527/6149316>.
- Shih-Tang Su, Vijay G. Subramanian, and Grant Schoenebeck. Bayesian Persuasion in Sequential Trials. In Michal Feldman, Hu Fu, and Inbal Talgam-Cohen (eds.), *Web and Internet Economics*, pp. 22–40, Cham, 2022. Springer International Publishing.

-
- Jeremy G Weber. Social learning and technology adoption: The case of coffee pruning in peru. *Agricultural Economics*, 43:73–84, 2012.
- Xupeng Wei and Achilleas Anastasopoulos. Social learning with a self-interested coordinator. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pp. 3137–3144. IEEE, 2022.
- Marcus Williams, Micah Carroll, Adhyyan Narang, Constantin Weisser, Brendan Murphy, and Anca Dragan. On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback, February 2025. URL <http://arxiv.org/abs/2411.02306>. arXiv:2411.02306 [cs].
- Jibang Wu, Zixuan Zhang, Zhe Feng, Zhaoran Wang, Zhuoran Yang, Michael I. Jordan, and Haifeng Xu. Sequential Information Design: Markov Persuasion Process and Its Efficient Reinforcement Learning, February 2022.
- Shuo Wu, Pawan Poojary, and Randall Berry. Observational Learning with a Budget, April 2025. URL <http://arxiv.org/abs/2504.19396>. arXiv:2504.19396 [cs].
- Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, Kathleen McKeown, and Tatsunori B. Hashimoto. Benchmarking Large Language Models for News Summarization. *Transactions of the Association for Computational Linguistics*, 12:39–57, January 2024a. ISSN 2307-387X. doi: 10.1162/tacl_a_00632. URL https://doi.org/10.1162/tacl_a_00632.
- Yadong Zhang, Shaoguang Mao, Tao Ge, Xun Wang, Adrian de Wynter, Yan Xia, Wenshan Wu, Ting Song, Man Lan, and Furu Wei. LLM as a Mastermind: A Survey of Strategic Reasoning with Large Language Models, April 2024b. URL <http://arxiv.org/abs/2404.01230>. arXiv:2404.01230 [cs].

A APPLICATIONS ELABORATED

This appendix grounds the two planner objectives of our framework (Section 3) in concrete applications, beginning with the altruistic case. Consider a sequence of individuals (agents, in our terminology) deciding whether to patronize a business or service provider (a restaurant, contractor, realtor, etc.), served information by a recommendation system (the planner). The unknown binary state is whether the business is good, and both the system and the agents want to match it: agents want to patronize the business if and only if it is good, and the system wants to induce exactly that. In this sense the recommendation system is an *altruistic planner*.

Here signal precision has a direct interpretation. The system can show an agent a highly targeted (*precise*) ad that showcases the business’s strengths and weaknesses in contexts she relates to, given her background, or a generic and confusing ad that does little to inform her. The precise signal is more informative, comprehensible, and relatable, and thus more likely to drive her to the correct conclusion – patronize if and only if the business is good – so social welfare rises as more agents decide correctly. But raising precision is costly, since it requires tailoring to the agent’s background, and the system must therefore choose precisions to maximize social welfare net of these costs. This is exactly the altruistic planner’s problem of Section 3.2.

We now turn to a real-world instance of a *biased planner*. Consider a group of voters (agents) motivated to support the candidate most likely to win in their community. This “back a winner” motive demonstrably shapes electoral outcomes – it is, for instance, why US states with early primaries exert outsized influence (see Bartels (1988)). The planner is a particular candidate’s campaign, which seeks to make agents back its candidate. The binary state is whether the candidate is winning, so an agent who knew the state would back the candidate if and only if he were winning.

Each agent understands her own community, so she could infer whether the candidate is winning (the state) if she correctly grasped his stances and policies – which she would, if that information were delivered in the form she digests best (audio-visuals for some, fact-laden long-form articles for others, brief focused content for others still, or simply her native language).

The campaign sends digital ads (private signals) of varying precision, where precision captures how well the content is tailored to the agent’s taste. Crucially, a precise signal conveys the true state faithfully – it helps the agent clearly understand the candidate’s policies, record, and character, letting her correctly infer whether he will win – but does *not* guarantee she backs him: if he is in fact losing in her community, a precise signal will turn her against him. An imprecise signal, by contrast, obfuscates the state (through vagueness or confusion) and thereby raises the chance she *mistakenly* thinks he is winning. This is why the biased planner is incentivized to *lower* precision when it believes its candidate is losing – the obfuscation regime our theory predicts in Section 5. Tailoring is costly in either direction: researching how an agent best absorbs content is required both to sharpen a message and to blunt it (e.g., knowing she prefers brief messages is what lets the planner confuse her with verbose ones). The planner thus chooses precisions to maximize the expected number of backers net of these tailoring costs.

B SOCIAL LEARNING DYNAMICS

We first summarize the sequence of developments at step i :

1. The planner determines q_i based on $\mathcal{H}_i$ and, q_i is observed by all agents.
2. Agent i receives private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i takes action a_i based on the observable history and their private signal with the aim of maximizing their own utility.
4. All other agents (and the planner) observe a_i and the public belief at the end of step i , b_{i+1} is updated accordingly.

Note that from (1), q_i is a function of $\mathcal{H}_i$. Thus, if a random variable is conditioned upon both q_i and $\mathcal{H}_i$, the conditioning on q_i is redundant. We will use this principle throughout.

B.1 PRIVATE BELIEF UPDATE

We now describe how agent i chooses action a_i . Let $\alpha_i := \frac{b_i}{1-b_i}$ be the public likelihood ratio based on observed actions. Agent i forms a private belief about ω , $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$, by combining the current public belief with her private signal, s_i . We consider a private likelihood ratio, $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. We now describe how $\tilde{\alpha}_i$ and $\tilde{b}_i$ are computed.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \quad (8)$$

$$= \frac{q_i}{1-q_i} \alpha_i \quad (9)$$

In obtaining Equation (8), we apply the assumption that, conditioned upon ω , agents' signals are independent of one another and the history to assert that $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$. The same is done when conditioning upon $\omega = B$. We additionally use the fact that q_i is a function $\mathcal{H}_i$.

2. If private signal is $s_i = B$, following the logic above,

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \quad (10)$$

We let $y(b, q)$ and $z(b, q)$ denote the probability of realizing signal G or B , with public belief b and signal precision q conditioned upon the history. By conditioning on ω and applying the Law of Total Probability, these probabilities can be expressed as follows:

$$\begin{aligned} y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\ &= q_i b_i + (1-q_i)(1-b_i) = 1 + 2q_i b_i - q_i - b_i \end{aligned} \quad (11)$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \quad (12)$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields the following:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1-q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \quad (13)$$

Note that if $s_i = G$ then $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i)$ cannot be 0. Similarly, if $s_i = B$, then $z(b_i, q_i)$ cannot be 0. Thus, the equation above is well-defined.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i 's action a_i will be G if $\tilde{b}_i$ exceeds 0.5 (equivalently, if $\tilde{\alpha}_i > 1$), B if $\tilde{b}_i < 0.5$ (equivalently, if $\tilde{\alpha}_i < 1$), and s_i otherwise as per the tie-breaking rule. This action and i 's precision q_i are observed by all peers, resulting in an updated public belief b_i .

When $b_i < 1 - q_i$, then $\alpha_i < \frac{1-q_i}{q_i}$. Therefore, if $s_i = G$, $\tilde{\alpha}_i < 1$ by Equation (9); if $s_i = B$, $\tilde{\alpha}_i < \frac{(1-q_i)^2}{q_i^2} \leq 1$ (since $q_i \geq 0.5$) by Equation (10). Thus, no matter the private signal realization s_i , $a_i = B$ when $b_i < 1 - q_i$. Similar arguments can be used to show that when $b_i > q_i$, $a_i = G$. Intuitively these cases are when the strength of the public belief overpowers the precision of the private signal.

Now consider the case that $1 - q_i \leq b_i \leq q_i$. Then if $s_i = G$, $\tilde{\alpha}_i \geq 1$ from Equation (8) and since $b_i \geq 1 - q_i$. If $s_i = B$, then $\tilde{\alpha}_i \leq 1$ from Equation (10) and since $b_i \leq q_i$. Together with the tie-breaking rule, this results in action $a_i = s_i$. This leads to Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Thus, agent i 's action is only informative in the first case of Equation (2) where it allows future agents to perfectly infer i 's private signal. Thus, in this case, public belief will update just as the private belief from Equation (13). In the latter two cases of Equation (2), the action contains no information about the true state. Thus, the public belief will not change. This yields Equation (3) reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (14)$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now prove that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, regardless of the rest of $\mathcal{H}_i$. It follows then that

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

From Equation (2), for the second and third cases, it is clear that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, regardless of the rest of $\mathcal{H}_i$. We consider the first case next.

$$\begin{aligned} \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) &= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first step applies the law of total probability. In the second step, using the fact that q_i is a function of $\mathcal{H}_i$, we substitute b_i for $\mathbb{P}(\omega = w | \mathcal{H}_i, q_i)$. In the third step, we use the fact that, for $1 - q_i \leq b_i \leq q_i$, $a_i = s_i$, and, further, that s_i is independent of the history when conditioned upon ω . The result follows.

This leads to the following theorem:

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G | b_i, q_i, a_i\} \quad (17)$$

In other words, b_i captures the entire information pertinent to the update of b_{i+1} that is contained in $\mathcal{H}_i$.

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G | b_i, q_i\}}{\mathbb{P}\{a_i | b_i, q_i\}}$$

We will prove the claim via induction on i .

For our base case, consider $i = 1$ so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem provides the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G|b_1, q_1\} \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\} \mathbb{P}\{\omega = G|b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \end{aligned}$$

where the final step follows as the unconditioned belief of ω is equal to the prior belief b_1 .

Now, assume the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G|\mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\} \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G|\mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\} \mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G|b_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n .

This completes the proof. $\square$

Note that we have shown that the public belief is Markovian adapting the arguments in Banerjee (1992) and Bikhchandani et al. (1992). The modifications were required because of the differences in selections of the agents' precisions caused by the introduction of the central planner.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We prove this for each case of a_i delineated in Equation (2). In the second case, we can reason as follows:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq G|b_i, q_i) = 1 - b_i$$

Similarly, in the third case of Equation (2), we obtain:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq B|b_i, q_i) = b_i$$

For the case when $1 - q_i \leq b_i \leq q_i$ and $a_i = s_i$ from Equation (2), we can reason as follows:

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqn 2} \quad (18)$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqns 11, 12, Thm 6} \quad (19)$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1-q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \quad \text{Eqn 13} \quad (20)$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases yields the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) allows us to reason as follows

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As stated in Section 3.2, we can restrict the altruistic planner to precision $q \in [p, 1]$ because social welfare is increasing in precision (see Appendix C.9). Accordingly, $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) leads to optimal precision p if $p < \max(b, 1 - b)$. That is, if the signal is not precise enough to influence the agent's action, then the planner will not expend resources for it. In the latter case, because $\beta(\cdot)$ is increasing and concave, $\dot{\beta}(\cdot)$ is positive and decreasing. Thus, $r_A(b, q)$, if the optimal lies in this range it must be attained at $q = 1$.

We now identify the regime in which the latter case is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, the above inequality is never satisfied, and it is never optimal to select precision 1. Otherwise, the inequality above yields $b \in \left[\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C} \right]$. This yields Theorem 1. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch Because convexity of the value function is the linchpin of our theoretical contribution, we sketch the argument before giving it in full. The proof is inductive: we show that the expected k -th stage reward – the planner's expected utility from the control and action of the k -th agent – is convex, with the instantaneous reward (Equation (5)), itself convex in the public belief, serving as the base case.

The first obstacle is the unusual public-belief update: although the state space is uncountably infinite, each update places support on at most two values. We handle this with a decision tree – the complete binary tree of all trajectories once an initial belief and policy are fixed – in which each node has two children, one per possible signal realization of the next agent. The root is the expected instantaneous reward at time 1 (the first agent's action), and the induction descends the tree, the k -th level holding 2^{k-1} nodes, each keyed to a sequence of $k - 1$ signal realizations.

The inductive step shows that if a node at level $k - 1$ has convex expected reward, so do its two children at level k . This is exactly where the action-belief coupling bites: two agents starting from different public beliefs may, under the same precision and signal, take opposite actions (see Equation (2)), so the standard tools for bounding future terms of a Markov process do not apply. What rescues the argument is a special feature of our update – Bayesian updates are martingales ($\mathbb{E}[b_{i+1}] = b_i$) – which, combined with the convexity of the instantaneous reward, closes the inductive step and thus the proof.

Preliminaries We first prove the following lemma containing a few useful properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1 - b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward function r_A . Note that the decision function of the agents when choosing their actions is symmetric from Equation (2), thus, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus, the instantaneous reward function is symmetric about 0.5.

Now consider the belief state update when beginning with belief n_i assuming signal precision q_i . we will use $\bar{x}$ and $\underline{x}$ to represent updated beliefs from x after observing action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \bar{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus, the belief updates are also symmetric. Combining this with the symmetry of the reward we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus, the expected rewards and belief state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities which we will denote as $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Thus, the branching probabilities of Λ_n and Λ_b are also symmetric, completing the proof.

Extremal values

We will show that $V_A^*(0) = 0$ and rely on the symmetry about 0.5 proven above to show that $V_A^*(1) = 0$.

From Equation (3), when $b_0 = 0$, $b_0^+ = b_0^- = 0$ for any value of precision. Thus, regardless of signal realizations or control actions, the agents' actions and, subsequently, the public belief, will remain unchanged. That is, $b_i = 0 \forall i$ and, from Equation (2), $a_i = B \forall i$.

From Equation (5), we can then write:

$$\begin{aligned}
V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right]
\end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We now introduce the concept of *decision trees* for Markovian and more general policies. A policy can be represented by a decision tree, which is a complete binary tree as in figure 3. Every node of the decision tree represents the state of the system, i.e., the corresponding belief value when the controller sends its private signal to the agent corresponding to the level of the node. The branches represent the actions taken by the agent corresponding to the level once it receives its private signal. The controller's policies (i.e. choice of precision) determine the probabilities of the actions from each node and the belief resulting from the action. The probabilities of an action at a certain node can be considered the weight of the branch. A decision tree is uniquely identified by the belief values associated with nodes and the weights of the branches. For a deterministic stationary Markovian policy, the controller's choice of precision, and therefore the probability associated with an action, is a deterministic function of the belief value of the node, regardless of the level of the node in the tree and the path to the node from the root. Thus, the decision tree of any such policy, i.e. the belief values at the nodes and the weights of the branches, is determined entirely by the value of the initial state. For an arbitrary policy of the controller, the controller's decision at a given node can depend on the entire path leading to the node.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We will first show that, for fixed precision, the instantaneous reward r_A is convex in public belief. We can rewrite r_A as follows using Lemma 7:

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave. As we are subtracting the minimum, the overall function is convex with respect to x .

We consider arbitrary x_0, m_0, n_0 such that $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. To show the convexity of $V_A^*(\cdot)$ we must show that $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. By definition of the optimal value function, $V_A^*(x) \geq V_A^\pi(x)$ for any policy π . Thus, it will suffice to show that there exist $\tilde{\pi}$ such that

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

We refer to the decision tree of π^* when the initial state is x_0 as Λ .

We consider $\tilde{\pi}$ to be a policy of the controller that chooses the precision at each node to be the same as what Λ does at the counterpart node. The nodes of decision trees are considered to be counterparts if they can be reached via the same sequence of actions from the root (see figure 3 for illustration). Thus, the precision chosen by this policy is the same as what Λ does after the same set of actions of the agents. We refer to the decision trees for $\tilde{\pi}$ with initial state m_0 (respectively, n_0) as Λ_m (Λ_n).

We now illustrate the policy $\tilde{\pi}$ for initial values m_0 and n_0 considering the decision trees Λ_m and Λ_n . While doing so, we illustrate counterpart nodes and the weights of the branches. We will select one path of actions λ out of these decision trees by specifying set $\mathcal{G}$ as the epochs where action G is taken and $\mathcal{B}$ as its complement. Refer to figure 3 for illustration. Considering the decision tree Λ , we will refer to x_i as the state at epoch i when beginning with belief x_0 and following π^* along this path of actions. That is, x_i assumes one of 2^i possible belief values at epoch i starting from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote the

public belief after beginning with m_0 (n_0), following the path λ . Node m_i (respectively n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i from decision tree Λ . Note that, $\tilde{\pi}(m_i)$ can now be illustrated notationally. Specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Similarly, $\tilde{\pi}$ can be defined at each node of Λ_m , considering the action taken by π^* at each counterpart node in Λ . Note that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m . This extends similarly to branches associated with action B and Λ_n . Thus, the three decision trees differ in both the values of the nodes and the weights of the branches. Since π^* is deterministic, so is $\tilde{\pi}$. Therefore, from Equation (13) and Equation (3), x_i , m_i , and n_i are deterministic quantities once the specific path of actions is specified by $\mathcal{G}$.

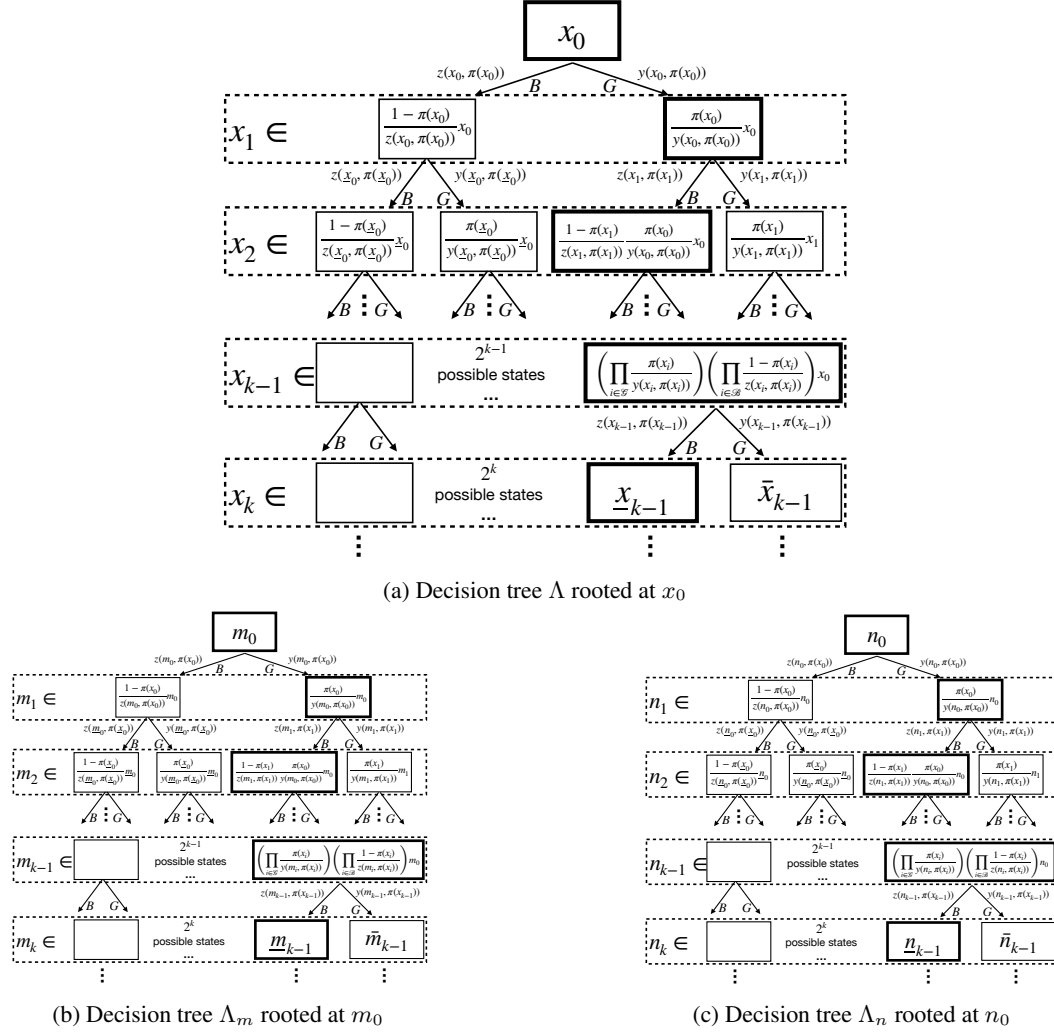


Figure 3: Here we depict the belief evolution in the form of three binary trees. The tree begins with state x_0 , m_0 , and n_0 , respectively, and level i contains 2^i possible states reachable after i epochs. Left branches correspond to action B leading to updated belief $\underline{x}_0$ (respectively, $\underline{m}_0$ and $\underline{n}_0$). Right branches correspond to action G leading to updated belief $\bar{x}_0$ (respectively, $\bar{m}_0$ and $\bar{n}_0$). Each branch is labeled with the corresponding probability, and the bold states indicate an example path, like λ . In the example path, $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. The bold states in each level of the three trees are counterparts.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ be the expected instantaneous reward at epoch k when beginning with state b_0 and applying policy π and $\phi^*(x) = \phi^{\pi^*}(x)$. Applying Fubini's Theorem, we can rewrite our value function

as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus, we can rewrite (24), as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

To prove this, we show that for all $i = 0, 1, 2, \dots$

$$\phi_i^*(x_0) \leq t \phi_i^{\tilde{\pi}}(m_0) + (1-t) \phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

leading directly to (26), (24), and finally the convexity of $V^*(\cdot)$ with respect to public belief.

Now let us consider general $i = k$. $\phi_{k-1}^\pi(x_0)$ will have 2^{k-1} terms each corresponding to one path of possible actions in the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ will yield the following term in $\phi_{k-1}^*(x_0)$

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, that is, the probability of following the specified path λ when beginning with state x_0 and following policy π_A^* . Again, we will use $\bar{x}_i$ and $\underline{x}_i$ to represent updated beliefs from x_i after observing action G or B , respectively. Thus, $\bar{x}_i$ and $\underline{x}_i$ constitute belief values of the children nodes (in the $(i+1)$ th level) of the node with belief value x_i in the i -th level. Refer to levels $k-1$ and k in figure 3a. Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (29)$$

Similarly, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, that is, the probability of following the specified path λ when beginning with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. We also define $P_\lambda^{\tilde{\pi}}(n_0)$ by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (31)$$

We will show that

$$\begin{aligned} & P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \leq t P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & + (1-t) P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \end{aligned} \quad (32)$$

Similarly, each pair of children for each term in the $(k-1)$ -th level of the belief tree can be combined and the corresponding inequalities can be obtained following the same process. Summing the resulting inequalities will yield (27), leading to (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now focus on the latter of the two terms of (29). We will show the following two properties

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned}
1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
&+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)
\end{aligned} \tag{34}$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \tag{35}$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \tag{36}$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-t) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \tag{37}$$

We apply this as follows to the latter term of (29):

$$\begin{aligned}
&P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\
&\leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\
&\quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\
&= t [P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Eqns 36, 37})
\end{aligned} \tag{38}$$

Applying the same argument to the first term of (29) yields:

$$\begin{aligned}
&P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\
&\leq t [P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1}))]
\end{aligned} \tag{39}$$

Summing (38) and (39) yields (32).

We now complete the proof of (32) and, therefore, the entire proof, by proving (33) and (34).

Proof of (33)

From the recurrence relation in Equation (13) and Equation (3), we can write $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as the product of all belief updates with x_0 , m_0 , and n_0 , respectively, as follows. Each term in the products corresponds to the multiplicative state update after a single action.

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \tag{40}$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \tag{41}$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \tag{42}$$

These equations may be best understood via the illustration in figure 3. We then reorganize terms to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, using the fact that $x_0 = tm_0 + (1-t)n_0$.

$$\begin{aligned}\underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \quad (43)\end{aligned}$$

This proves (33).

Proof of (34)

We first consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We now prove that this can be written as $\alpha_0 m_0 + \beta_0$ for some α_0, β_0 which depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$ and the nature of the dependence is determined by the sequence of actions along this path in stages 1 to $k-1$. We will prove this via induction working backwards from the final term (i.e. the $(k-1)$ -th term) of the product. Therefore our base case will be the final term of this product which can be either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$, depending on the preceding action (in the $(k-1)$ -th stage).

$$\begin{aligned}y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1}))\end{aligned}$$

Thus, the final term of the product is of the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$ where α_{k-1} and β_{k-1} depend only upon $\pi_A^*(x_{k-1})$, and the nature of the dependence is determined entirely by the $(k-1)$ -th action (since this action determines whether the final term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, assume that $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for some $\alpha_{j+1}, \beta_{j+1}$ which depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the nature of the dependence is determined entirely by the sequence of actions in this path for $i > j$ to $i = k-1$. We will now show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_j m_j + \beta_j$ again with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ with the nature of the dependence determined by the sequence of actions on

this path for $i > j-1$ to $i = k-1$. We define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$.

Thus,

$$\begin{aligned}&\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1})\end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, by our inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions in this path from $i > j$ to $i = k-1$. Thus, the claim holds for α_j, β_j , completing the induction.

Now recall that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$. Note that the policy and sequence of actions is the same as those for the numerator. Thus, because, as shown, the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the path of actions, the same logic can be applied to write the denominator

as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 . This can be done similarly for the numerator of the second term in (34). Thus, we can reason about the coefficients in (33) as follows:

$$\begin{aligned}
& t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\
& = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = t m_0 + (1-t) n_0) \\
& = 1
\end{aligned}$$

Thus, (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Assume by way of contradiction that there exists $b \in [0, 1]$ such that $\pi_A^0(b) > \pi_A^*(b)$. We will show that $V_A^*(b) < V_A^\pi(b)$ for a policy π we construct, violating the optimality of $V_A^*(\cdot)$.

Let π be such that, when starting at public belief b , the policy π applies precision $\pi_A^0(b)$ at the current time step and then applies the optimal policy at all future time steps. We now elaborate $V_A^*(b) - V_A^\pi(b)$ by breaking the value function into instantaneous reward r_A and future discounted reward. We will refer to b^+ and b^- as the updated beliefs after receiving signals G and B , respectively when applying the optimal policy and b_m^+ and b_m^- as the same when applying the myopic policy.

$$\begin{aligned}
V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\
&+ \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\
&< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\
&\leq 0 \quad (\text{Theorem 2})
\end{aligned}$$

Here, we applied the fact that the instantaneous reward under $\pi_A^0(\cdot)$ must be greater from the definition of the myopic optimal policy.

We then relied on the convexity of $V_A^*(\cdot)$ from Theorem 2 for the future cost. From Equation (13), note that $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$. Along with this convexity, this leads to the last step above and completes the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ with policy π applying precision q in the current time step and applying the optimal policy at all future time steps. The q that maximizes $V_A^\pi(b)$ provides the optimum precision starting at public belief b . Because the planner never benefits from decreasing precision below the baseline p (see Appendix C.9), we restrict q to $[p, 1]$ without loss of generality.

We will let b^+ and b^- denote the positive and negative belief updates possible after receiving a signal with precision q . Taking the second derivative of $V_A^\pi(b)$ with respect to q for $q \geq \max(b, 1-b)$:

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1-b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1-b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

With the convexity and concavity of $V_A^*(\cdot)$ and $\beta(\cdot)$, respectively, the above shows that $V_A^\pi(b)$ is convex with respect to q on $[\max(b, 1-b), 1]$. Thus, on this interval, it is maximized at one of the two extreme points.

For $q \in [0.5, \max(b, 1 - b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative; thus, the optimal choice will be the baseline precision p to minimize the cost. Thus, there are three possible optimal precisions: p , $\max(b, 1 - b)$, and 1.

Note that the expected value under the latter two candidates is always strictly negative. Under precision p , however, the expected value is 0 at $b = 0$ and $b = 1$. This, combined with the non-positivity and symmetry of the value function (Lemma 8) implies the existence of d_A such that $\pi_A^*(b) = p$ for $b \in [0, d_A] \cup (1 - d_A, 1]$. Furthermore, from Theorem 1 and applying Lemma 9, $d_A \in (0, t_M)$ where t_M is defined as in the statement of Theorem 1.

Finally, Theorem 1 and applying Lemma 9, also imply the existence of $t_A \in [d_A, t_M]$ such that $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$. This completes the proof. ■

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Applying the action rule from (2), the instantaneous reward (6) can be written as follows:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Note that $z(b, q) = b + q - 2bq$ increases with respect to q when $b < 0.5$ and decreases with respect to q when $b > 0.5$.

At $b = 0.5$, $z(b, q) = z(0.5, q) = 0.5$ regardless of the precision chosen. Furthermore, at this public belief, $q \geq \max(b, 1 - b)$ because $q \geq 0.5$ for any q , falling in the first case of (44). In this case, the second term of $r_B(b, q)$ is unchanging with respect to q , so the planner seeks to maximize only the first term, which is accomplished at $q = p$.

We now prove a series of lemmas (10-14) which will assist us in our proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except $q = p$. Thus, we can write its derivative with respect to the chosen precision as follows (except at $q = p$):

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, we can write the second derivative of the reward with respect to q as follows, for $q \neq p$:

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Because $\beta(\cdot)$ is concave, this expression is non-negative, and $r_B(b, q)$ is convex with respect to q . Thus, the reward is maximized at one of the extreme points $\{\max(b, 1 - b), 1\}$ or p because $r_B(b, q)$ is not differentiable at $q = p$. □

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Because $p > 0.5$, for $b \in [0, 1 - p)$, $\max(b, 1 - b) = 1 - b > p$.

First, consider the case when $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. From (44), in such instances, the reward is maximized by minimizing $|q - p|$. Thus, the optimal precision lower than $\max(b, 1 - b)$ is p .

Now consider $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$. What remains is to rule out $\pi_B^0(b) = 1$. To do so, we compare $r_B(b, 1)$ and $r_B(b, 1 - b)$. Both fall under the first case of (44). Because $\beta(\cdot)$ is increasing in its argument and $p < 1 - b \leq 1$, the first term of the reward from (44) is greater for precision $1 - b$ than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$ so we can rule out precision 1. This completes the proof. □

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. The reward is strictly negative for any $q \neq p$ because of the term $-\beta(|q - p|)$. For $b \in (p, 1]$ and $q = p$, $p < \max(b, 1 - b) = b$, leading to the third case of (44) and yielding reward 0. Thus, the optimal precision is p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ contains only $\{0.5\}$. At $b = 0.5$ $p = 1 - b$, thus $\pi_B^0(b) = p = 1 - b = 0.5$. We deal with $p > 0.5 \geq b$ for the remainder of the proof.

For $b \in [1 - p, 0.5]$, $\max(b, 1 - b) = 1 - b \leq p$. Thus, from the first case of (44), precision p yields reward $r_B(b, p) = -Cz(b, p)$. We now consider two cases: $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

Suppose $q < \max(b, 1 - b) = 1 - b$. This leads to the second case of (44) and yields reward $-\beta(|q - p|) - C$. Furthermore, $z(b, q)$ is increasing with respect to q for $b < 0.5$ and takes values between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$. Thus, $z(b, q) < 1$ for all q . Therefore, the reward from policy $q < 1 - b$ is strictly lower than $r_B(b, p)$, and we can rule out this case.

When $q \geq \max(b, 1 - b)$, applying Lemma 10 yields that the remaining candidate precisions are $\{1 - b, p, 1\}$. We rule out precision 1 by comparing it to precision p . The first term of the reward from (44) is greater for precision p than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, the second term is also greater for precision p , and we can rule out precision 1. This completes the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

Note that from the definition, any optimal policy is also ϵ -optimal for any $\epsilon > 0$. Thus, $\pi_B^0(b)$ is also ϵ -optimal

Following this definition, let $\pi_B^{\epsilon,0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or

2. $\pi_B^{\epsilon,0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, $\max(b, 1 - b) = b \leq p$. We consider two cases: $q \geq \max(b, 1 - b)$ or $q < \max(b, 1 - b)$.

In the first case, Lemma 10 yields that $\pi_B^0(b) \in \{b, p, 1\}$. We rule out precision b by comparing it to precision p . The first terms of the reward from (44) is greater for precision p . Furthermore, since $z(b, q)$ is decreasing with respect to q for $b > 0.5$, the second term of the reward is also greater for precision p . Thus, we can rule out precision b . This yields the first claim.

In the latter case, $q < \max(b, 1 - b) = b$. Therefore, (1) since $q \geq 0.5$, $q \in [0.5, b)$, and (2) the third case of (44) applies leading to reward $-\beta(|q - p|)$. Thus, the reward $r_B(b, q)$ is an increasing function of $q \in [0.5, b)$ because $0.5 < b \leq p$. This leads directly to the second claim. $\square$

Remark 3. As noted in Puterman (1990) Section 2.3.1, an optimal policy may not exist for infinite action spaces because the supremum may not be attained. When an optimal policy does not exist, we instead seek ϵ -optimal policies, i.e., policies that yield reward within $\epsilon > 0$ of the supremum. In particular, this is true for case 2 of Lemma 14.

Proof of Theorem 4 We consider the myopic biased policy in 4 cases: $b \in [0, 1 - p]$, $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$, making use of the fact that $p \geq 0.5$. Based on Lemmas 11-14, these cases have the following candidate policies: $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$, respectively.

Case 1: $b \in [0, 1 - p]$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We will show the existence of a threshold t_1 such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

We can express the reward under the two candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

The derivative of $r_B(b, 1 - b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

which is positive as $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Furthermore, when evaluated at $b = 0$ and $b = 1 - p$ we obtain

$$\begin{aligned} r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

Thus, $r_B(b, 1 - b)$ is increasing on $[0, 1 - p]$ with $r_B(b, 1 - b) < r_B(b, p)$ for $b = 0$ and vice versa for $b = 1 - p$. Because $\beta(\cdot)$ is continuous, so are both reward functions. Thus, there exists $t_1 \in (0, 1 - p)$ such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

If $p = 0.5$, this interval contains only $\{0.5\}$, and, for $b = 0.5$, $1 - b = p$. Thus, the two candidate policies are equivalent, and the optimal policy is $\pi_B^0(b) = p$. For the remainder of this case, we consider $p > 0.5$.

We now prove the existence of t_2, t_3 such that $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3]$, and $\pi_B^0(b) = p$ for $b \in [1 - p, t_2] \cup [t_3, 0.5]$.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This leads to the following condition for the optimal policy to be $1 - b$:

$$\begin{aligned} r_B(b, 1 - b) &\geq r_B(b, p) \\ \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1 - p, 0.5]$. The RHS of (46) is a concave parabola that increases in the first half of $[1 - p, 0.5]$ and decreases in the second half.

Now consider the two halves of the interval $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. In the first half, the LHS and RHS of (46) are increasing and concave, with the RHS also being a parabola. Thus, in the first half, the two intersect at most two points (one of which is at $b = 1 - p$). In the second half, the LHS of (46) is increasing while the RHS is decreasing. Thus, they intersect at most once. Let the two possible intersections for $b > 1 - p$ be t_2 and t_3 with $t_2 \leq t_3$. Thus, what remains is to determine the optimal on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1 - b)$ because $p > 0.5$, thus $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. Subsequently, $\pi_B^0(b) = 1 - b$ for (t_2, t_3) , and $\pi_B^0(b) = p$ for $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

In this case, we show the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, and, therefore, $b - \epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now, note that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions in b while $r_B(b, b - \epsilon)$ is convex in b (from the concavity of β). Thus, $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and, as a result, have convex level sets. This implies that the set on which both of these expressions are positive, i.e., the set for which $b - \epsilon$ is the optimal policy, must also be convex. Therefore, $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

What remains is to compare precisions 1 and p on the interval $(0.5, t_5]$. To do so, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 and their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Because $r_B(0.5, 1) < r_B(0.5, p)$ there exists interval $(0.5, t_4)$ with $t_4 \leq t_5$ such that $\pi_B^0(b) = p$. Furthermore, because both are linear in b , there is at most one crossover point where their ordering changes. Thus, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows directly from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward with respect to b can be written as follows:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Because $q \geq 0.5$ and $z(b, q) \in [0, 1]$, it follows that $r_B(b, q)$ is increasing with respect to b .

Next, note that the Bayesian nature of belief updates implies that the belief process is a martingale. That is, for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To demonstrate this consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$, and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the former case, it follows from (3) that $b_{i+1} = b_i$, satisfying the claim.

In the latter case, (13) gives the distribution of b_{i+1} which yields the same.

The monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first prove that the optimal value function is continuous in the following lemma:

Lemma 16. Biased value function continuity

$$V_B^*(\cdot) \text{ is continuous w.r.t. } b.$$

Proof. First consider the one-stage value function defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$. Thus, we can equivalently write the one-stage value function as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1-b, 1, b-\epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

Using the above expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. First, consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1-b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of

(44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum, $r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_\epsilon < p$ such that on $[t_\epsilon, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to public belief and the fact that the control variable lies in the compact set $[0.5, 1]$, one might conclude that the optimal is guaranteed to exist at all public beliefs. However, there is an important subtlety which prevents such a conclusion: while $V_B^*(\cdot)$ is continuous with respect to public belief, it need not be continuous with respect to precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)); thus, $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted as $\phi(b_i, q_i)$. Now, b_{i+1} need not be continuous with respect to q_i , thus $\phi(\cdot)$ may not be continuous with respect to q_i . Thus, there may be regimes in which the optimal value is not attained by any policy. Indeed, we will show that such cases exist.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. We again let $z(b, q) = b + q - 2bq$ be the probability of an agent receiving signal B with precision q and prior public belief b and $y(b, q) = 1 - z(b, q)$ be the corresponding probability of the agent receiving signal G . We abuse notation to allow $V_B(b, q)$ to be the expected utility when beginning at public belief b , applying precision q in the current time step, and then following the optimal policy $\pi_B^*(\cdot)$ in all future steps. We can express $V_B(b, q)$ as follows by applying (13), (3), and (44):

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ + \delta y(b, q) V_B^*(f(b, q, G)) \\ + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ -C\mathbf{1}(b < 0.5) \\ + \delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49) when $q < \max(b, 1 - b)$. Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as close to p as possible. Thus, if the optimal precision is less than $\max(b, 1 - b)$, then it will be as close to p as possible.

We now consider four cases: (1) $b \in [0, 1 - p)$, (2) $b \in [1 - p, 0.5)$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p)$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p)$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p)$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1-b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q-p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1-b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1-\epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q-p|) \rightarrow \beta(1-p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1-p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p-p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1-p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1-p, 0.5)$, $\pi^*(b_i) \geq 1-b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1-b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1-b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1-\delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1-\delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1-b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1-b_i$.

The remaining candidate policies are $q_i \in [1-b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1-b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1-b_i)$ and $b \in [1-p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1-q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1-b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1-b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the

supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b)} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \tag{57}$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q)V_B^*(b, q) \\ &\quad + \delta z(b, q)V_B^*(b, q)] \end{aligned} \tag{58}$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. It is clear that the one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Furthermore, note that the convexity of social welfare with respect to public belief follows from the same argument as in the proof of Theorem 2. This is because social welfare is equivalent to the altruistic value function in the special case where $\beta(\cdot) = 0$.

We can then follow the same argument as in the proof of Lemma 9 (Appendix C.4) to show that increasing precision increases welfare. $\square$

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

As a first step toward the heterogeneous-agent generalization flagged in Section 7, we show here that, for the altruistic planner, our problem is equivalent to one with two types of individuals holding opposing preferences.

Assume that there are two types of agents G and B with the type of agent i denoted t_i . The first type receives unit reward when $a_i = \omega$ and no reward otherwise. The second receives unit reward when $a_i \neq \omega$ and no reward otherwise. Further, assume that all agents know their own type and the types of all other agents. All other aspects of the model remain the same.

The private belief update of agent i can be expressed as below. This remains unchanged because it only depends on the signal structure, i.e., $\mathbb{P}(s_i = \omega) = q_i$.

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_i q_i - b_i - q_i} b_i & s_i = G \\ \frac{1-q}{b_i + q_i - 2b_i q_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), agent i 's action can be written as follows with superscript indicating the type of agent i and $!$ indicating negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because we assume the type of each agent is known to all, when $q_i \geq \max(b_i, 1 - b_i)$, all agents can perfectly infer the private signal of agent i . If $q_i < \max(b_i, 1 - b_i)$, the action of agent i is uninformative.

Thus, after observing agent i 's type t_i , precision q_i , and action a_i , the public belief is updated as follows:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this setting, an altruistic planner seeks to induce type- G agents to take action ω and type- B agents to take action $!\omega$ as these choices align with the personal utility functions of the agents.

Thus, we can express the planner's instantaneous reward as follows with superscript denoting the type of agent i :

$$\begin{aligned} r^G(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega) \\ &= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases} \\ r^B(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega) \\ &= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases} \end{aligned}$$

These two rewards are identical to each other and to the reward in Equation (5), which establishes that the altruistic planner's problem is unchanged in the presence of heterogeneous but observable agent types. $\blacksquare$

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

This appendix details the setup behind our empirical contribution. To simulate controlled social learning, we consider a sequence of agents deciding whether to buy a new car model, each informed by observed past actions

and an ad whose precision the planner selects. LLMs fill three distinct roles – agents, planner, and oracle – and the planner’s and agents’ information sets and objectives match the model of Section 3 exactly. Each agent is assigned a ‘profile’ of demographic information together with car use cases and preferences, and at the onset we designate either the good or the bad car as the offering.

The oracle plays a two-fold role: it generates a generic message to serve as the baseline-precision signal absent planner intervention, and, once the planner chooses a precision, it generates a message of that precision customized to the target agent’s profile. To do so it is given a fact sheet (safety rating, customer satisfaction, etc.) for each car type alongside the agent profiles; together these let it produce a tailored message at the planner’s prescribed precision. Further detail on setup and prompting appears in Appendix E.2.

The remaining two roles mirror the model directly. Each agent makes a binary decision from the observed history of prior decisions and its private signal, aiming to match the true state. The planner sees the same decision history and chooses a precision level for each agent’s message, having also been given the precision of a generic, un-tailored message it may send for free; its objective is to induce correct decisions in the altruistic case and to induce purchases in the biased case.

For a ground-truth comparison, we solve the planner’s MDP analytically via JuliaPOMDP (Egorov et al., 2017) and evaluate the LLM simulations against it.

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

All experiments use the LG AI EXAONE Deep 32B model (LG AI Research, 2025), and we solve the model analytically and run the LLM simulations across the parameter grid in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	{0.1, 0.3, 0.5, 0.7, 0.9}
p	Baseline signal precision	{0.6, 0.7, 0.8, 0.9}
δ	Discount Factor	{0, 0.25, 0.5, 0.75, 1}

Table 1: Instance parameters swept in both the analytical and LLM experiments.

We designed the car facts so that the good car is good for everyone and the bad car is bad for everyone, removing heterogeneity in the underlying state as a confound. The fact sheets for the GOOD and BAD cars, respectively, are reproduced below.

```
fact_sheet["GOOD"] = [
    "EPA-rated range: 290 miles.",
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",
    "8-year / 100,000-mile battery warranty.",
    "NHTSA 5-Star overall safety rating.",
    "Cold-soak -5 °F keeps 93 %% range.",
    "Cools cabin 100→72 °F in 3 min.",
    "Cargo volume 26 ft³ total.",
    "Tow's 1 500 lb with factory hitch.",
    "Average 1.2 service visits first 3 yrs.",
    "92 %% owners would buy again (Apr-25).",
    "Qualifies for federal credit.",
    "Quarterly OTA updates incl. battery pre-heat."
]
fact_sheet["BAD"] = [
    "EPA-rated range: 155 mi (combined).",
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
    "Battery warranty: 3 yr / 36 000 mi, first owner only.",
    "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
    "Cold-soak 15 °F retains 58 %% range; cabin heat cuts range 20%.",
    "Cabin cools 105 °F → 72 °F in 11 min.",
    "Cargo volume 17 ft³; rear seats fixed (no fold-flat).",
    "No factory towing approval.",
    "Average 3.8 service visits per year (battery-alert issues).",
    'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
```

```

1890         "Not eligible for federal tax credit.",
1891         "OTA updates limited to infotainment."
1892     ]
1893 
```

Profiles were generated programmatically, each specifying an agent's name, age, city of residence, interests, planned use case for the car (e.g., commuting, road trips), and major pain points or concerns (e.g., fuel efficiency, winter driving).

1897 E.2.2 PROMPTS

1898 *Agent LLM Prompt:*

```

1899
1900 sys = (
1901     f"You are deciding whether to buy a car. "
1902     f"Your profile is {profile}"
1903     f"You start believing there's a 50% chance the car is GOOD. "
1904     f"You have two information sources: "
1905     f"1. Previous agents' decisions "
1906     f"2. An advisor's message about the car "
1907     f"Your goal is to maximize your reward "
1908     f"i.e. BUY if the car is GOOD and "
1909     f"NOT BUY if the car is BAD. "
1910     f"If you BUY a GOOD car, you GAIN ${C}. "
1911     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1912     f"If you BUY a BAD car, you LOSE ${C}. "
1913     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1914     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1915     f"decision = exactly 'BUY' or 'NOT BUY'."
1916 )
1917 user = (
1918     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1919     f"ADVISOR MESSAGE:\n{message}\n\n"
1920     f"TASK: Use the previous decisions and the advisor message. "
1921     f"Decide BUY or NOT BUY. "
1922     f"Update your belief about car quality (0.0-1.0). "\n"
1923     f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}'
1924 )
1925 
```

1922 *Planner LLM Prompt:*

```

1923
1924 sys = (
1925     f"You are providing messages to potential car buyers. "
1926     f"Each potential car buyer (or agent) has the same information. "
1927     f"Your goal is to get agents to {objective}."
1928     f"Each decision according to your objective, earns you {C} dollars. "
1929     f"You and the agents have the same information. "
1930     f"Initially there is a 50% chance the car is GOOD. "
1931     f"The agents' aim to BUY if the car is GOOD "
1932     f"and NOT BUY if the car is BAD. "
1933     f"You can send a baseline message for free "
1934     f"or tailor the message to the agent at a cost. "
1935     f"Tailoring costs {k} dollars times the change in precision. "
1936     f"(Message precision is the probability (0.5 to 1.0) "
1937     f"that the message leaves the buyer with the correct impression. "
1938     f"For example, if the car is likely GOOD: "
1939     f"a high precision message will likely suggest BUY "
1940     f"and a low precision message will likely suggest NOT BUY. "
1941     f"You may choose to increase or decrease precision. "
1942     f"Agents decide based on your message and the decision history. "
1943     f"They update beliefs by observing previous decisions. "
1944     f"IMPORTANT: Thus your objective is to maximize:"
1945     f"{C} * (# of agents who choose according to your objective) "
1946     f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1947     f"where {d} is your discount factor for future cost. "
1948 )
1949 
```

```

1944 user = (
1945     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1946     f"DECISION HISTORY:\n{history_txt}\n"
1947     f"TASK: "
1948     f"Based on the history and your objective, "
1949     f"choose your desired precision.\n"
1950     f'JSON response: {{"message_precision": <number>}}\n'
1951 )
1952
1953 Oracle LLM Prompt:
1954
1955 sys = (
1956     f"You will be given "
1957     f"1. an agent profile, "
1958     f"2. information about a car, "
1959     f"3. the true quality of the car, "
1960     f"4. a baseline message and the 'precision' of that message, and "
1961     f"5. a desired precision for the message you output. "
1962     f"(Message precision is the probability (0.5 to 1.0) "
1963     f"that the message leaves the buyer with the correct impression. "
1964     f"For example, if the car is likely GOOD: "
1965     f"a high precision message will likely suggest BUY "
1966     f"and a low precision message will likely suggest NOT BUY. "
1967     f"IMPORTANT: Output a message with the requested precision. "
1968     f"IMPORTANT: The message should be tailored to the profile. "
1969 )
1970 user = (
1971     f"AGENT PROFILE:{profile}\n"
1972     f"TRUE CAR QUALITY: {omega}\n\n"
1973     f"CAR FACTS: {car_facts}\n"
1974     f"BASELINE MESSAGE:{baseline_message}}\n"
1975     f"BASELINE MESSAGE precision: {p}\n"
1976     f"DESIRED MESSAGE precision: {q}\n"
1977     f"IMPORTANT: Output a message with the requested precision. "
1978     f"TASK: Write a message (MAX 100 words) with the desired precision."
1979     f'JSON response: {{"message": "<message>}}\n'
1980 )

```

1978 E.3 VALIDATION OF LLM BELIEFS AND ORACLE

1981 To ensure our empirical findings rest on sound measurements, this appendix reports two validation experi-
1982 ments: one for the self-reported LLM beliefs used in figure 1b, and one for the LLM oracle’s fidelity. Treating
1983 the LLMs as rational, utility-maximizing, but not necessarily Bayesian agents, we aim to (1) show that a
1984 model’s self-reported posterior genuinely governs its choices in state-contingent decisions, and (2) verify that
1985 the oracle’s messages actually carry the requested precision.

1986 In the first experiment, we give the LLM a prior and a new signal, record its self-reported posterior, and then
1987 ask for a binary “buy/not-buy” decision that pays unit reward if and only if the true state is good. A rational
1988 agent should choose “buy” exactly when its belief exceeds 0.5, and the LLM’s decisions track this threshold
1989 sharply (figure 4a): it overwhelmingly chooses “buy” when its reported posterior exceeds 0.5 and “not buy”
1990 otherwise. The second experiment elicits the model’s willingness to pay for a state-contingent payoff: for a
1991 range of prices c , we ask whether the agent would pay c in exchange for unit utility if the state is good. A
1992 rational agent with belief p should accept exactly when $c < p$. Figure 4b plots the fraction of “buy” decisions
1993 across posterior–price pairs and shows the predicted separation – acceptance is high when $c < p$ and low
1994 when $c > p$ – precisely as expected from a utility-maximizer whose belief is p . Together, the two experiments
1995 confirm that the self-reported posterior closely matches the belief the model actually uses when deciding.

1996 Finally, we validate the oracle’s ability to craft messages of a target precision. After the oracle gener-
1997 ates a message of specified precision q , we ask a separate LLM to estimate its precision $\hat{q}$, sweeping
 $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ over 100 programmatically generated profiles with 10 trials per combina-
tion – 6,000 trials in total. The mean absolute error $|q - \hat{q}|$ was just 0.025, confirming that the oracle’s realized
precision closely follows the requested precision and that our precision control is faithful.

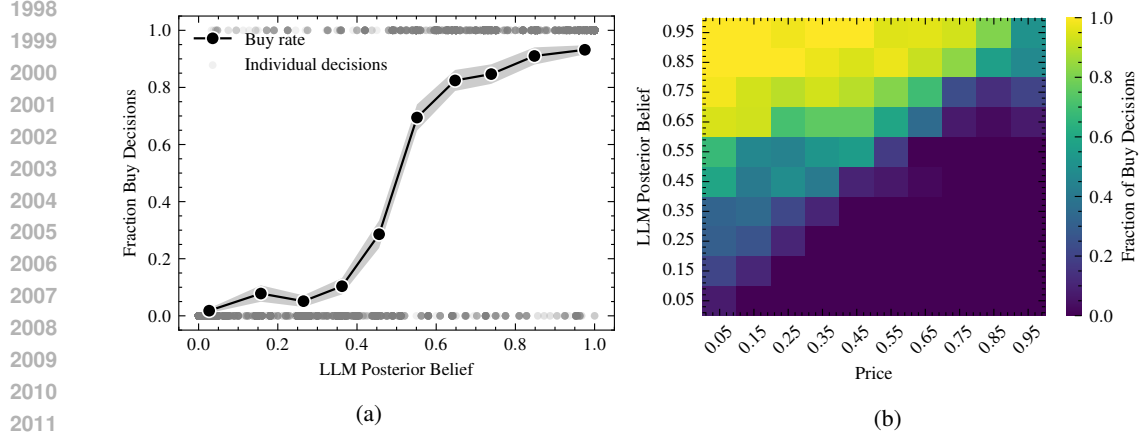


Figure 4: Validation that self-reported beliefs drive decisions: the rate at which the LLM chose a ‘buy’ decision as a function of (a) its self-reported posterior belief, and (b) its self-reported posterior together with the price of buying.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

To connect the micro-level biases of Section 6.1 to their macro-level consequences, we unpack belief trajectories from an example LLM instance in which we fix the policy to the analytically optimal one and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0, whereas the LLM belief (solid black line) is markedly more “stubborn.” This aggregate trajectory follows directly from the individual biases: the public belief resists entering a cascade (NB3) because agents overreact to the occasional positive signal that cuts against the group’s increasingly negative belief (NB2).

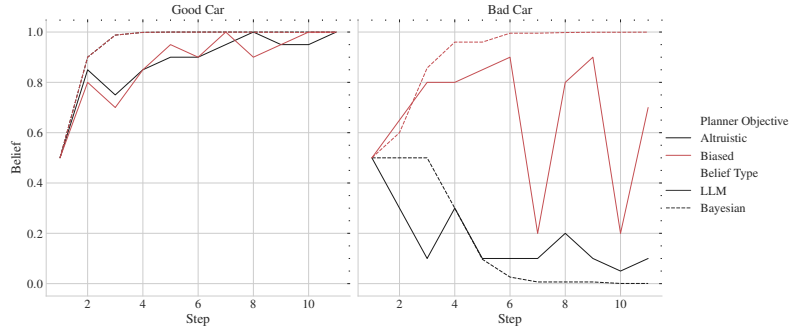


Figure 5: Agent belief-updating trajectories – the true Bayesian belief versus the belief reported by the LLM agent – under the analytically optimal policy, across planner objectives and true states, illustrating how the LLM’s non-Bayesian biases slow convergence to the true state.